



MSAR Box (Medication Storage and Reminding)

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Pacharapol
Areelak**

**Suwattanarat
Duangrat
Viriyakiat
Sukkaew**

**BACHELOR OF ENGINEERING
INFORMATION AND COMPUTER ENGINEERING**

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A COMPUTER ENGINEERING PROJECT SUBMITTED TO
MAE FAH LUANG UNIVERSITY IN PARTIAL FULFILLMENT OF
THE REQUIREMENTS FOR THE DEGREE OF
BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING

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THIS COMPUTER ENGINEERING PROJECT HAS BEEN APPROVED
TO BE A PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE DEGREE OF BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING 2021

EXAMINING COMMITTEE



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ACKNOWLEDGEMENTS

Foremost, we would like to express our sincere gratitude to our precious advisor Asst. Prof. Dr. Suppakarn Chansareewittaya for supporting our project and giving the opportunity for us to do our research and developing this project. Without valuable guidance, this project might not succeed.

With all support from our parent with encouragement and fortune is propelling us to our successful were very grateful and we felt very appreciative.

The support of the equipment from the Multimedia Laboratory at Mae Fah Luang University were leading us to the goal.

Lastly, we would like to thanks any committee and any suggestion from everyone from helping us improve the quality of our work and made this project in the best state possible.

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Abstract

MSAR Box Project is designed to Help doctors and patients with treatment. The MSAR Box can help you contact the doctor via the “Video conference” with the camera and the screen on top of the box. So, patients can talk and check the symptoms in their home no need to go to the hospital or clinic. Appointment can be made through the website by the screen attached to the box. Also the MSAR Box can help remind you to take the medicine on time with the alarm sound by speaker and the light from the LED that attached to the front of the box.

This project we have prepared the acrylic to be the main body of the box, Raspberry Pi 3 b+ to be the main controller of every function in the box such as the video conference, the sound remainder, the LED light and the electric door lock. We hope that the users that have been using our project in daily life will have a more comfortable life so they don't need to contact the doctor in the hospital so often and your health will be better because of the efficiency of the medicine that has been taken on time.

Keywords : Pill timer, Medicine reminder, Telemedicine

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CHAPTER 1

INTRODUCTION

1.1 Background and Rational

Due to Thailand society nowadays tends to become an aging society, many problems caused by this situation are mainly the medical treatment in the household. Many households with elder tend to have a problem with the medicine given by the doctor due to many variations of medicine that have different specific amounts of time to consume. Also, the appointment with the doctor is quite problematic due to traffic and work time of other household members which might result in a transportation issue. Nowadays, with the occurrence of virus COVID-19 outbreak, Having the appointment face to face with the doctor has become much more problematic.

Then we decided to develop the Alarm med box for this purpose. This piece of work comes with features solely for this problem such as alerting the patient or any household member to reminding them a time for specific medicine given by the hospital and collecting data for confirming the medicine intake. The Camera and Speaker were also connected for having video conference to the doctor right at the household which resolve the transportation problem for patient with mildly symptom and hard to transport

1.2 Objective

- 1.2.1 To develop equipment for helping elders with memory problems.
- 1.2.2 Making an appointment with a doctor becomes less problematic for elders.

1.3 Scope

- 1.3.1 The box can contain medicines.
- 1.3.2 Elder take medication at appropriate time.
- 1.3.3 The doctor can take teleconferences at any time.

1.4 Methodology

- 1.4.1 Proposal preparation and defense
- 1.4.2 Literature review and requirement gathering
- 1.4.3 System analysis
- 1.4.4 System design
- 1.4.5 Senior project 1 document preparation
- 1.4.6 Senior Project 1 exam
- 1.4.7 System development
- 1.4.8 System testing
- 1.4.9 Senior Project 2 exam
- 1.4.10 Senior Project 2 exam

1.5 Plan

Table 1.1 Working plan

	Pre-project				
Plan/Month	Jan	Feb	Mar	Apr	May
Proposal preparation and defense					
Literature review and requirement gathering					
System analysis					
System design					
Senior Project 1 presentation					

	Project						
Plan/Month	Feb	Mar	Apr	May	Jun	July	Aug
System development							
System testing							
Senior Project 2 presentation							

1.6 Expected Result

- 1.6.1 MSAR Box were developed.
- 1.6.2 This developed smart teleconference medicine box has the potential to join the start up competition.
- 1.6.3 The doctor can contact the user with higher efficiency and less effort.

1.7 Place and Equipment

1.7.1 Place

- 1.7.1.1 Dormitory
- 1.7.1.2 Home
- 1.7.1.3 MI Space room

1.7.2 Hardware

- 1.7.2.1 Acrylic sheets
- 1.7.2.2 Raspberry Pi 3
- 1.7.2.3 SD card
- 1.7.2.4 Camera
- 1.7.2.5 Microphone
- 1.7.2.6 Speaker
- 1.7.2.7 Magnetic lock
- 1.7.2.8 Monitor screen

1.7.3 Software

- 1.7.3.1 Raspberry Pi OS

CHAPTER 2

LITERATURE REVIEW

2.1 Related Research

1. Suparpit von Bormann, Winai Sayorwan, Sirichai Channim, Sararat Rungruangkhanarak, Premchai Suksamran, Piyaporn Srisuk, From College of Nursing, Nonthaburi Kanchanabhishek Institute of Medical and Public Health Technology King Mongkut's University of Technology North Bangkok has research about the medicine reminder with the bag. We can use the research to apply with our project
[website <http://ojslib3.buu.in.th/index.php/health/article/view/5780>]

2. AKKANONT PONGLUXSAMANA has researched telecommunication in many operating systems. we can use it to apply with our project.
[website http://www.sure.su.ac.th/xmlui/bitstream/id/df70d19f-9cb6-4492-a5e1-0567a4c9ed7a/Akkanont_Pongluxsamana_fulltext.pdf?attempt=2]

2.2 Related Work

1. Application Alive powered by AIA This application can Telemedicine and use Video conference with the doctor and can make appointments to contact the doctor also it has many features due to the AIA company.
2. Application Raksa This application can help you to contact the doctor every 24 hour and it can detect which medicine that you need and the medicine will be sent by the delivery.
3. PillDrill It is a product that is used to remind you to take the medicine. the product will have the box to store the medicine and the time alarm.

2.3 Related Technology

1. Zoom , Google meet, Webex is one of the related Technologies that use in many platform in the telemedicine industry



figure 2.3(1) showing technologies used in telemedicine

2. Raspberry pi is the technology that can be used to control many kinds of things with the electric circuit. It was very useful in our project.



figure 2.3(2) raspberry pi 3

2.3.1 Software

1. Visual Studio Code

Visual Studio Code is a source-code editor made by Microsoft for Windows, Linux and macOS. Features include support for debugging, syntax highlighting, intelligent code completion, snippets, code refactoring, and embedded Git. Users can change the theme, keyboard shortcuts, preferences, and install extensions that add additional functionality. Visual Studio Code is a source-code editor that can be used with a variety of programming languages, including Java, JavaScript, Go, Node.js, Python and C++. It is based on the Electron framework, which is used to develop Node.js Web applications that run on the Blink layout engine. Visual Studio Code can be extended via extensions, available through a central repository. This includes additions to the editor and language support. A notable feature is the ability to create extensions that add support for new languages, themes, and debuggers, perform static code analysis, and add code linters using the Language Server Protocol.

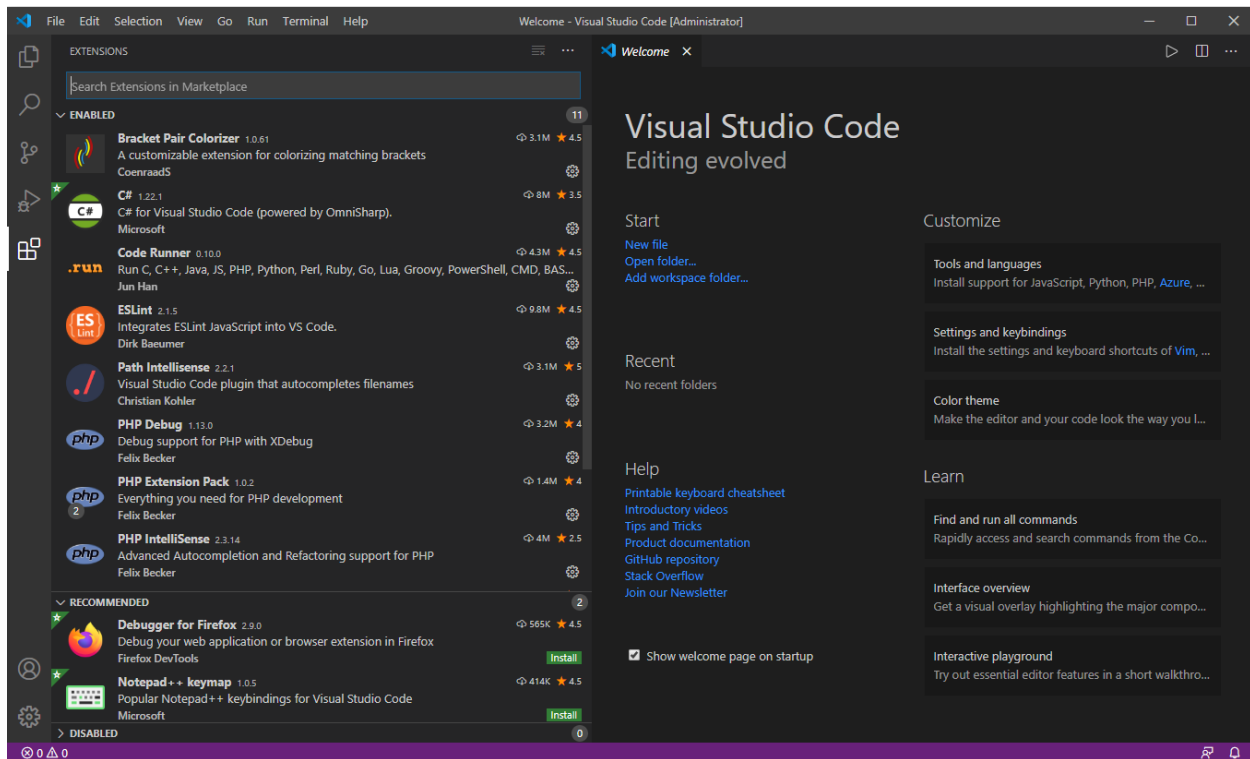


figure 2.3.1.1 Visual Studio Code

2.3.2 Language and tool development

1. Xampp

is a tool that help simulate web server we use to simulate web for some content that we need to show in simulate our website before run actual run

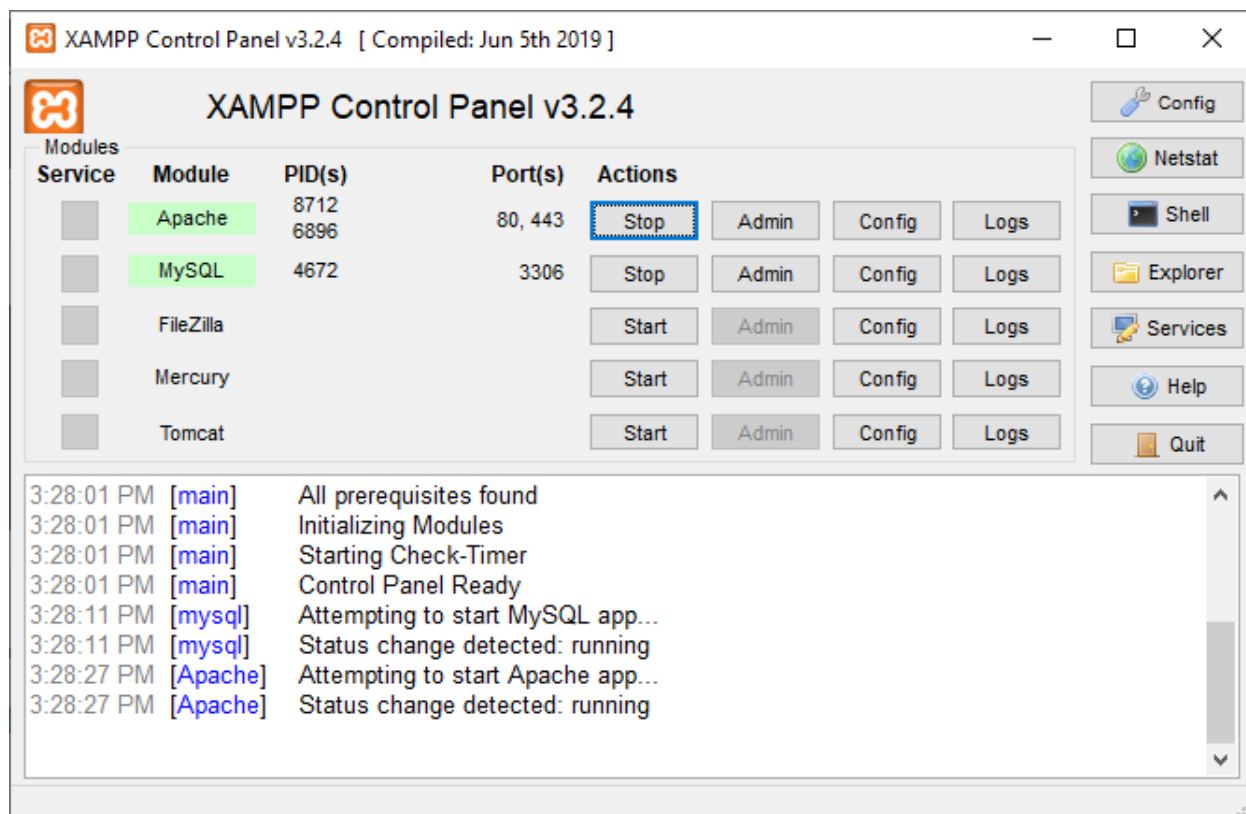


figure 2.3.2.1 Xampp interface

2.JavaScript

is a programming language that help combine features create web pages that can make website have animated. Javascript is programming language that can update and change html and css

```

    }).then(response => {
      setProgress('finished');
    });
  }

  return (
    <div className={`progress-button ${progress}`}>
      <span className="loading-text">loading</span>
      <button className="download-button">download</button>
      <span className="button-text">progress</span>
    </div>
  );
}

```

figure 2.3.2.2 Javascript showing code

3.ReactJS

is a front-end javascript that is free for building interface users on UI components .ReactJS can be use based in development like single-page or mobile



```

import React from 'react';
import ReactDOM from 'react-dom';

class ReactJSOverview extends React.Component {
  render() {
    return (
      <div>
        ReactJS facilitates the building of responsive user interfaces,
        able to perform well in highly scalable environment
      </div>
    );
  }
}

```

figure 2.3.2.3 ReactJS showing code

4.NodeJS

is a back-end javascript that is free for use cross platform for server to run web server by write command line tools and server-side scripting-running scripts server-side to produce dynamic web page content in server before sending to users web .

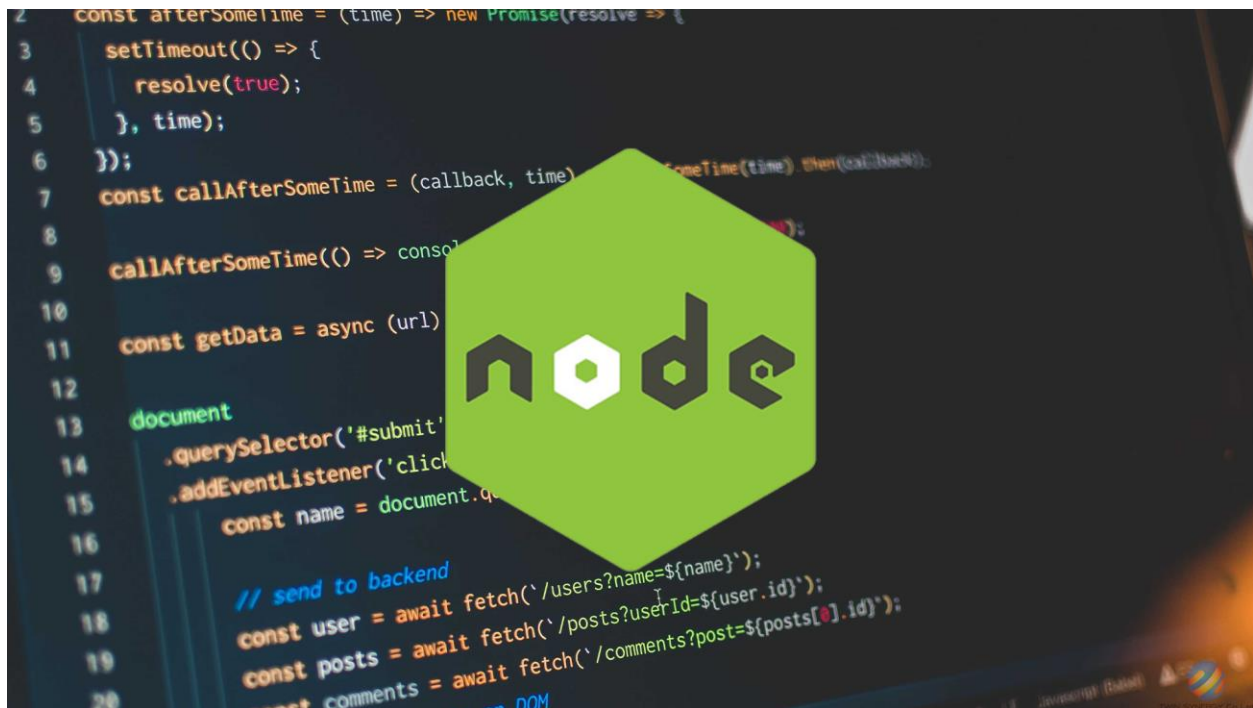


figure 2.3.2.4 showing code of nodeJS

CHAPTER 3

ANALYSIS AND DESIGN

3.1 Existing System

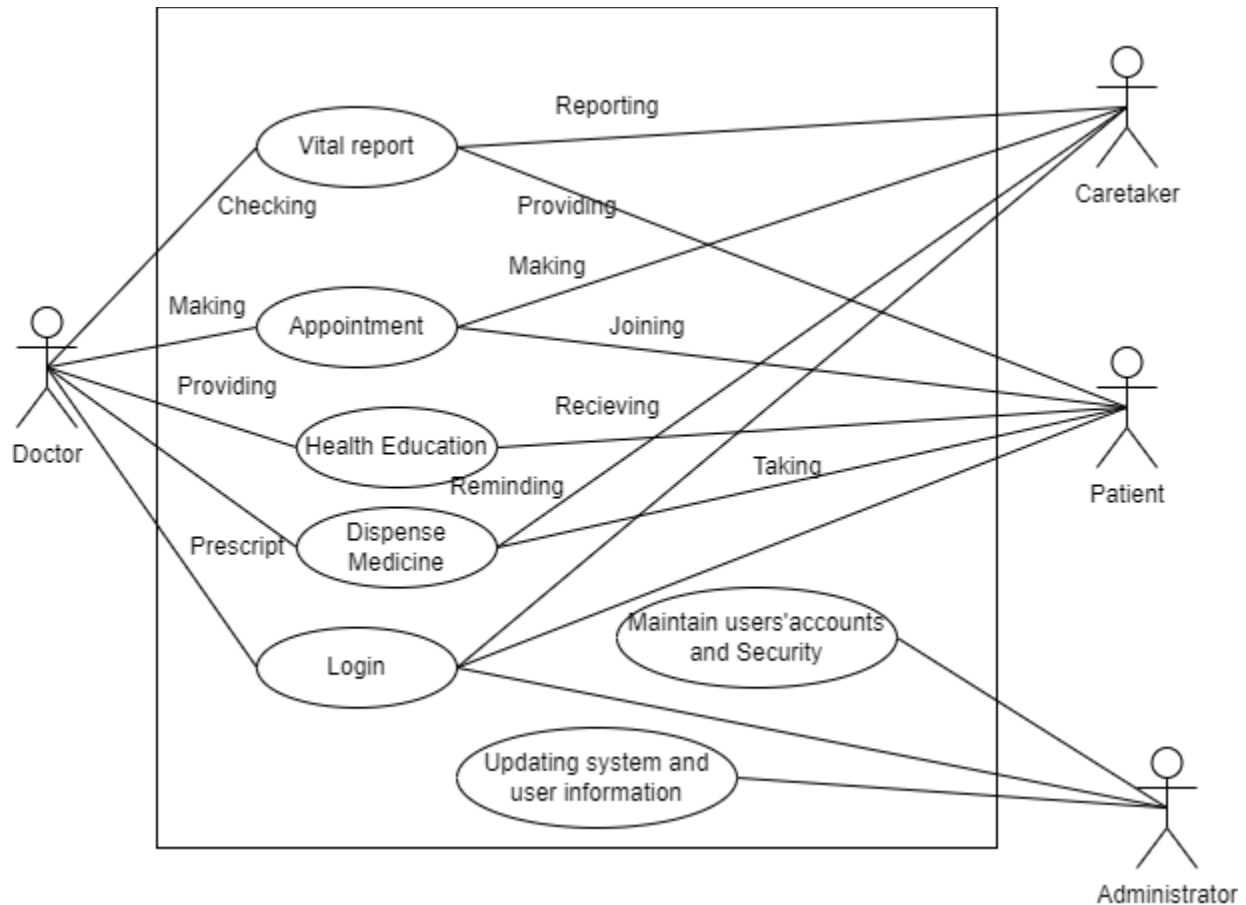


Figure 3.1: Showing existing system use case diagram

This use case diagram shows a system that was commonly used in many telemedicine applications, Focused on providing basic healthcare services. This system was designed to be used online due to the patient's current situation or virus pandemic. Even though this design is commonly used it's still causing problems to some caretakers or patients that do not have much knowledge on technology.

3.2 New System

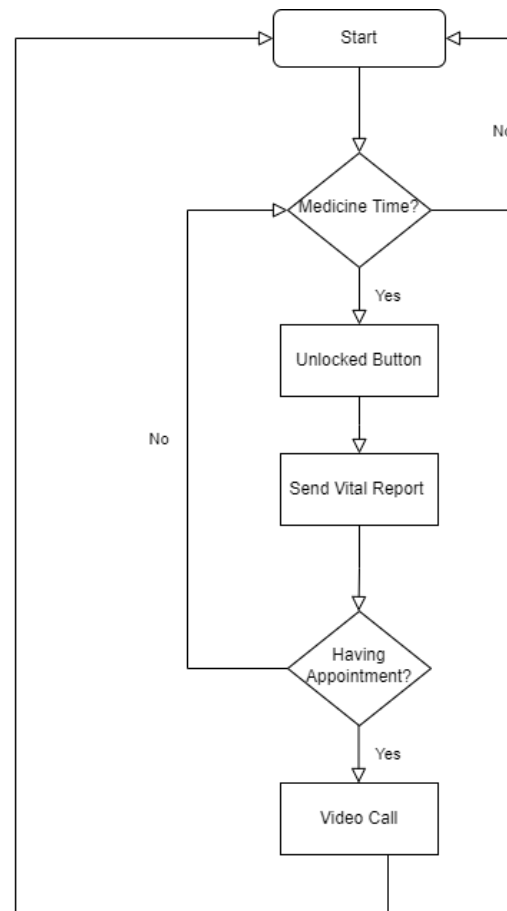


Figure 3.2: Showing our new system flowchart

Our project intended to create a tool for helping many patients contacting doctors and reminding the user to take medication right on time with simplicity. This flowchart showed how our device worked as our purpose.

3.3 Use case diagram

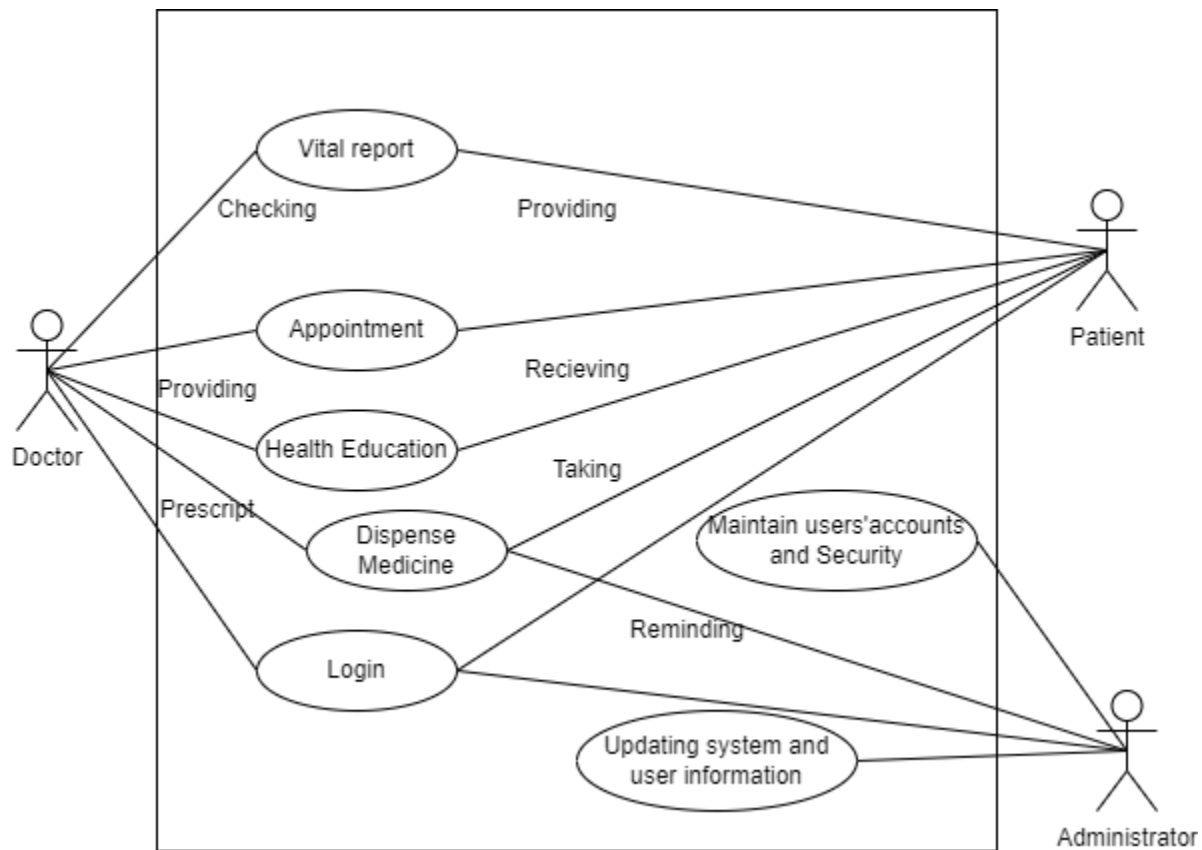


Figure 3.3: Showing our use case diagram

Our project has lessened many complicated issues from the existing system that is currently used. Making patients be the main user without requiring any help from caretakers is our solution. The device provides services that often need caretakers such as making video calls or reminding medicine intake time.

3.4 Activity diagram

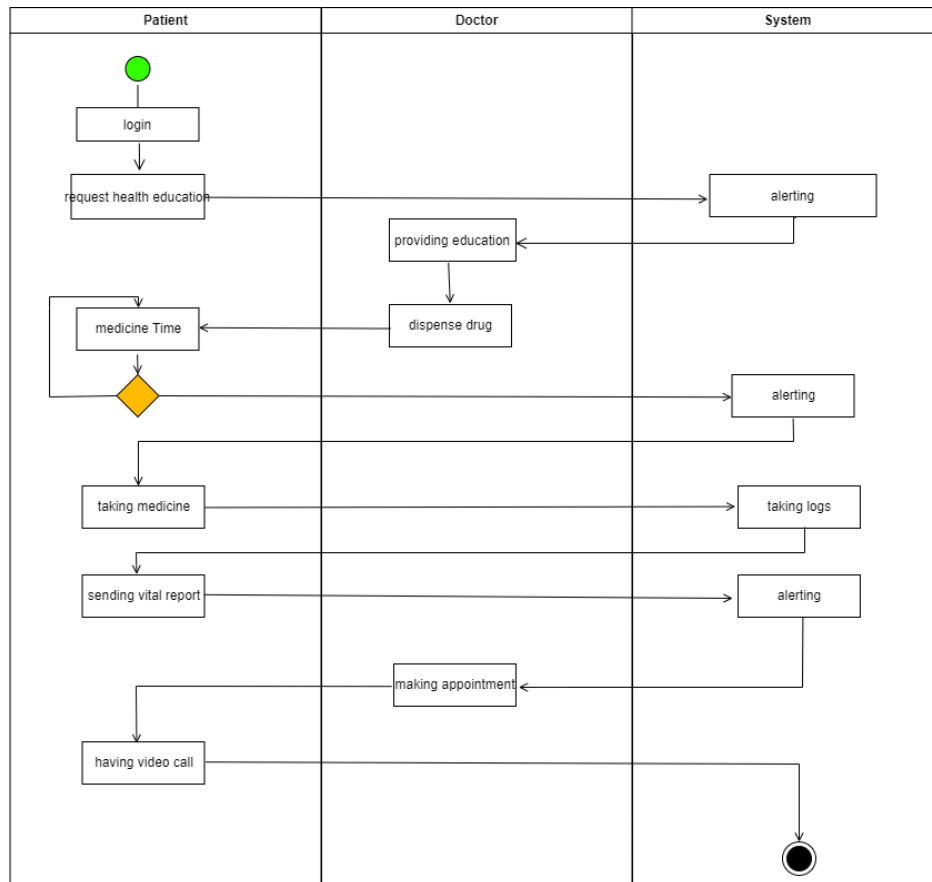


Figure 3.4: Showing our activity diagram

This activity diagram shows how the device works in detail. Starting from turning on the device to logging in and then we proceed through the process until the user shutting down a device.

CHAPTER 4

CONCLUSION

4.1 Conclusion

We have researched many products and information to develop our product and also we have the experience from the Startup Thailand Ledge 2022 to help us with more knowledge of marketing that can develop our project in the future. Now we have developed the hardware which is the box body and developed the software technology which is the operating system on Raspberry pi. We have tested some of the telemedicine systems but it still needs more development to be more useful. Our website we have developed and have tested some of it but it still needs more development with the more creative design and more colorful.

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4. https://en.wikipedia.org/wiki/Visual_Studio_Code
5. <https://www.educba.com/what-is-xampp/>
6. https://www.simplilearn.com/tutorials/reactjs-tutorial/what-is-reactjs#what_is_react
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